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Social Separation in Monkeys

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Phenomena associated with social separation from attachment objects in nonhuman primates are comprehensively reviewed. A biphasic protest–despair reaction to social separation is often seen in monkeys, as in human children. However, upon reunion there is generally a temporary increase in attachment behaviors rather than a temporary phase of detachment, as has been reported in the human literature. Gross factors such as age and sex do not appear to influence the responses to separation or reunion substantially. Rather, behavioral repertoires prior to separation and the nature of the separation and reunion environments appear to be more important determinants of the severity of separation reactions. These findings are consistent with the human literature. The evidence bearing on possible long-term consequences of early separations is also discussed. Finally, four different theoretical treatments of separation phenomena are presented and evaluated in light of existing data: Bowlby's attachment-object-loss theory, Kaufman's conservation–withdrawal theory, Seligman's learned helplessness theory, and Solomon and Corbit's opponent-process theory.

The study of social separation from objects of attachment has received widespread attention since early reports of the potentially devastating consequences that such separations can have for young children (e.g., Bowlby, 1960, 1973; Robertson & Bowlby, 1952; Spitz, 1946). Extensive research efforts have followed these initial studies, employing a variety of species. In most of these studies, *social separation* has operationally involved *the restriction of access to an individual (or individuals) with whom a subject had previously established a social relationship, however loosely defined.*

Interest in social separation has taken a number of directions. Some investigators have used a subject's response to separation as an index of the strength of its attachment bond to a given social object (e.g., Cairns, 1966). Others have studied separation as a means to uncover mechanisms underlying social development (e.g., Hofer, 1975) or to exaggerate the dynamics of existing social relationships (e.g., Hinde & Spencer–Booth, 1967). Recently a good deal of interest has centered on the use of separation as a procedure to induce experimental psychopathology—in particular, models of anaclitic depression (see Suomi & Harlow, 1977, for a review). In the following review we focus primarily on the latter two areas of interest: how studies of social separation have aided in the understanding of both the dynamics of social relationships and the etiology of anaclitic depression. Throughout, our emphasis is on studies of separation performed on nonhuman primates, primarily macaque monkeys.

We begin by reviewing the common characteristics of responses to separation and delineating the complex set of interacting variables that appear to determine how a given individual will respond to separation from a given object of attachment. It is the study of the determinants of this response to separation per se that has been of greatest interest to those working with models of anaclitic depression. We then turn to an examination

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of the persisting effects that separation may have even after reunion with the attachment object has occurred. The study of behavior following reunion has been of particular interest to those pursuing the dynamics and development of social relationships. Finally, we examine the evidence bearing on questions about long-term effects of separation. In particular, how long do the effects of separation persist, in what ways do any long-term effects manifest themselves, and to what extent does early experience with separation predispose an organism to react severely to other separations that may occur later in life? Following an examination of these three facets of social separation phenomena, we turn to an evaluation of four prominent, yet very different theoretical perspectives on separation—Bowlby's (1973) ethological attachment-object-loss viewpoint, Kaufman's (1973, Kaufman and Rosenblum, 1967) conservation–withdrawal interpretation, Seligman's (1974, 1975) learned helplessness theory, and Solomon and Corbit's (1974) opponent-process approach.

The Protest–Despair Response to Separation in Monkeys

Nature of the Protest–Despair Biphasic Response

The early work of Spitz (1946), Robertson and Bowlby (1952), and Bowlby (1960) indicated that young human children often display a biphasic response to prolonged physical separation from their mothers. Immediately after the act of separation the children tended to be highly agitated and distressed, a phase that Bowlby termed *protest.* This phase was typically followed, after several days, by a phase of dejection and social withdrawal termed *despair.* In these initial reports there was little evidence of recovery from the despair until reunion occurred.

The first systematic study of mother–infant separation performed with rhesus monkeys (*Macaca mulatta*) confirmed that Bowlby's biphasic response is not unique to humans. Seay, Hansen, and Harlow (1962) separated 6-month-old rhesus monkeys from their mothers for a 3-week period, during which the separated mother–infant pairs could see and hear but not contact each other. Seay et al. found that the infants initially responded to separation with excessive vocalizations, disoriented locomotion, and repeated attempts to break through the Plexiglas barriers separating them from their mothers. Within a few days, however, the infants' behavior changed considerably, and they became lethargic and socially withdrawn relative to their preseparation behaviors. These two phases corresponded very closely to Bowlby's protest–despair response seen in human children. At about the same time, Jensen and Tolman (1962) also reported the occurrence of vigorous protest in 5–7-month-old pigtail macaque infants (*Macaca nemestrina*) when they were separated from their mothers for short (less than 1 hour) periods.

Since these early investigations, several other primate studies have confirmed the occurrence of protest–despair responses when infants are separated from their mothers. For example, Hinde, Spencer–Booth, and Bruce (1966) raised rhesus infants in pens containing mothers, peers, and an adult male for the first 6 months of life. When the mothers were then removed from the pen for a 6-day period, clear-cut analogs to protest and despair were observed. Kaufman and Rosenblum (1967) removed the mothers of pigtail infants from a group pen for 28 days and also recorded biphasic reactions (*agitation, depression*), with the infants exhibiting some evidence of "spontaneous" recovery by the third week of separation. In both studies, protest or agitation was characterized by hyperactivity and excessive vocalizations, whereas despair or depression was characterized by sharp decreases in play and other social activities, as well as by increases in passive, self-directed behavior.

Recent evidence has also shown that the protest–despair reaction can occur in feral environments as well as in the laboratory. Singh (1975) observed the behavior of four infant rhesus monkeys before, during, and after the removal of each of their mothers from their troop. All four infants (two from town-dwelling and two from forest-dwelling troops) showed initial agitation upon removal of the mother, followed by a phase of markedly reduced play interaction with other animals. In addition, Goodall (cited in Hamburg,

Hamburg, & Barachas, 1973) reported similar reactions in chimpanzees (*Pan troglodytes*) living in the wild.

A majority of the studies investigating mother–infant separation in monkeys have employed systematic sociobehavioral observations as the only dependent variables. Recently, however, several research groups have begun to explore physiological and psychophysiological concomitants of the protest–despair response. For example, Reite and his co-workers at the University of Colorado Medical Center (Reite, Kaufman, Pauley, & Stynes, 1974; Short & Reite, 1977) have monitored electroencephalographic (EEG), electrocardiographic (EKG), and electrooculographic (EOG) activity, body temperature, and muscle activity prior to, during, and following maternal separation. To date, they have found that during protest, heart rate and body temperature become elevated and that REM sleep shows a sharp decrease on the first night. By the second day, most infants display decreases in body temperature and heart rate from preseparation levels, although intersubject variability on these measures increases markedly.

At the University of Wisconsin—Madison, some preliminary studies have examined the biochemical correlates of behavioral reactions to separation. Breese and his colleagues (1973) separated 6-month-old infants from their mothers for 6 days and then sacrificed the subjects; control monkeys were sacrificed immediately upon separation. Behaviorally speaking, the experimental subjects exhibited protest prior to sacrifice, while controls did not. The biochemical data indicated that relative to the controls, the experimental subjects showed significant elevation of tyrosine hydroxylase, dopamine-*B*-hydroxylase, phenylethanolamine-*n*-methyl transferase, and choline acetylase in the adrenal tissue, as well as elevated tyrosine hydroxylase and choline acetylase in the superior cervical ganglion of the sympathetic nervous system and serotonin in the hypothalamus. These findings thus revealed that monkeys exhibiting behavioral protest are also undergoing extreme physiological stress. Comparable data for correlates of the despair phase have not yet been published.

More recently, Levine and his co-workers at Stanford University have measured adrenocortical changes that occur in infant squirrel monkeys (*Saimiri sciureus*) when separated from their mothers for brief periods of time, for example, less than 1 day. Hunt, Smotherman, Lowe, McGinnis, and Levine (1977) reported sharp increases in cortisol levels 3 hours following separation. However, the cortisol levels were not significantly correlated with the behavioral reactions of the monkeys (see also Levine, Mendoza, Coe, Smotherman, & Kaplan, 1977).

Overall, these findings suggest that separation from its mother is physiologically stressful for an infant monkey, although the severity of its behavioral reactions may not correlate with all of the physiological stress indices. It is as yet unclear to what extent these stress reactions are unique to social separation or similar to reactions caused by other known stressors.

Other Reactions to Separation in Primates

Having described the prototypical protest–despair response to maternal separation, we must emphasize that this biphasic reaction does not always occur in either human or nonhuman primates. Rosenblum and Kaufman (1968) first emphasized this fact when they reported that bonnet macaque infants (*Macaca radiata*) were scarcely affected by separations from their mothers. These infants effectively transferred their attachment behaviors to other adults remaining in their social group and showed no indication of depression. More recently, attention has been drawn to the fact that, even within an individual species, there exists considerable variation in the form, severity, and duration of reaction to separation. For example, Lewis, Young, McKinney, and Kraemer (1976) compared the results of five different studies of mother–infant separation in rhesus monkeys. The studies were quite similar with respect to design, experimental procedures, and methods of data collection. However, there was enormous variation in the results. Only one of the studies found a consistent, clear-cut protest–despair response across all subjects. The variability in reaction could not be attributed to any

procedural differences among the studies. Similar variation in response to separation was found long ago in human primates. In Spitz's (1946) original study, only about one fourth of his population of separated infants exhibited the syndrome.

Clearly these results indicate that there is no universal response to maternal separation. Rather, it seems likely that numerous variables are operating, and probably interacting, to determine any given individual's response to separation. We now turn to a consideration of variables that are presently known to influence response to maternal separation.

Factors Influencing Response to Maternal Separation

Species. As mentioned earlier, a majority of the studies of mother–infant separation in primates have used rhesus or pigtail macaque monkeys, both of which frequently show protest–despair responses to separation. With respect to other species, Schlottmann and Seay (1972) separated 7-8-month-old Java monkey infants (*Macaca fasicularis*) from their mothers for 3 weeks and observed considerable despair, although there was only mild protest. Studies with patas monkeys (*Presbytis entellus*) (Preston, Baker, & Seay, 1970) and with squirrel monkeys (Kaplan, 1970; Jones & Clark, 1973) demonstrated protest but minimal, if any, despair. Rosenblum and Kaufman (1968) reported only minimal protest and a clear-cut failure to observe any despair in their study of bonnet macaque infants. This does not mean, however, that bonnet monkeys are incapable of experiencing biphasic reactions to social separation. Recently, Kaufman and Stynes (1978) reported a protest–despair reaction in one bonnet infant that was left in a group of pigtail mothers and infants after separation. (See discussion of separation environment below for an elaboration of this point.)

Overall it is probably true that different monkey species vary in their propensity to react severely to social separation. However, it is not yet clear to what extent these differences are primary or secondary to other species differences in social and maternal behavior. In recent years a number of investigators have argued in favor of the latter alternative (Hinde, 1976; Kaufman & Rosenblum, 1969; Suomi & Harlow, 1977).

Age and sex. Age of the infant at the time of separation has varied considerably in studies on young rhesus monkey subjects. Within the range of 2 months (Suomi, Collins, & Harlow, 1973) to 9 months (Rosevear, 1975), age at the time of separation appears to be a relatively unimportant factor (e.g., Hinde & Spencer–Booth, 1971). An exception to this generalization occurred in the study of Suomi et al. (1973), who found that 90-day-old infants exhibited qualitatively more severe reactions than monkeys separated at 60 or 120 days of age. This finding was attributed to the fact that the fear response becomes fully mature in rhesus infants by 90 days of age (Sackett, 1966; Suomi & Harlow, 1976); hence, infants of this age are particularly sensitive to any major change in their environment. Actually, the small effect of age over the 2–9-month range is quite striking in light of the major changes in the mother–infant relationship that occur during this time (e.g., Hansen, 1966; H. F. Harlow, M. K. Harlow, & Hansen, 1963; Hinde & White, 1974).

Similarly, among subjects under 1 year of age, there appear to be few major sex differences in reaction to separation from the mother. Hinde and Spencer–Booth (1971) do suggest that males may be slightly more adversely affected than females, but the differences are hardly overwhelming. Other researchers seldom mention sex differences, certainly a sign that whatever differences do exist are likely to be minor in nature. Again, this is somewhat surprising given the large sex differences in other aspects of the social development of macaque monkeys that consistently emerge in the first year of life (H. F. Harlow & Lauersdorf, 1974; Sackett & Ruppenthal, 1975). Sex differences in response to separation have been reported in human subjects, with males usually exhibiting the more adverse reactions (Bowlby, 1973).

Duration and number of separations. As has been reported in the human literature, long periods of separation from the mother can be more deleterious for young monkeys than short separations, but only if the separation environment is impoverished. Suomi et al.

(1973) found that rhesus infants permanently separated from their mothers and housed in single cages showed severe signs of despair and did not recover. By contrast, infants separated from their mothers but housed in pairs showed less severe signs of despair and generally exhibited considerable recovery over the next few months. Kaufman and Rosenblum (1967) also found evidence for some "spontaneous" recovery following maternal separation in group-housed pigtail infants after about 3 weeks. Therefore, duration of separation appears to interact with the nature of the infant's environment during separation in determining the severity of a given infant's reaction.

There have been very few studies published to date investigating the effects of more than one separation on the behavioral responses of monkey subjects. (See below for a discussion of multiple peer separations.) Spencer–Booth and Hinde (1971) found that reaction to a second 6-day separation was approximately the same as for a 6-day separation performed 1 or 2 months earlier. Chappell and Meier (1975) also reported that rhesus infants subjected to four 6-day separations showed few changes in response to each succeeding separation.

It is apparent that gross factors such as species, age, sex, and duration or frequency of separations do not account for all of the variance seen in response to maternal separation. We now turn to two factors that do appear to have a more substantial influence on responses to separation.

Preseparation history. An important source of variation in response to maternal separation across infants lies in differences that exist in their social repertoires before separation. In the human literature there is compelling evidence that differences in response to separation can often be traced to differential preseparation patterns of social responding (Bowlby, 1973; Heinicke & Westheimer, 1966). In particular, anxiously attached infants are far more likely to exhibit a severe reaction to separation than securely attached infants. *Anxious attachment* has been described by Bowlby as a syndrome existing when the infant has no confidence that his or her attachment object will remain available and accessible and hence tends to maintain close proximity in order to ensure continued access. *Secure attachment* can be said to exist when no such constant apprehension about separation is present. This variable appears to be important because securely attached infants have generally spent more time away from their mothers and as a result may have had more actual experience forming social relationships with other individuals prior to separation than anxiously attached infants have had.

Data from several monkey studies appear to corroborate the human findings. For example, Kaufman and Rosenblum (1967) noted that the one pigtail infant in their original study that failed to exhibit a despair response to separation happened to be the offspring of the dominant female in the group, and as a result it had spent more time than the other infants interacting with other adult females before separation. In addition, Kaufman and Rosenblum (1969) have argued that the dramatic species differences between bonnet and pigtail macaques are in large part traceable to differences in rearing practices and hence also in functional behavioral repertoires. Although pigtail mothers are generally possessive and seldom permit their infants to interact with others in their social group that are not members of the immediate family, bonnet mothers are more permissive and freely allow their infants to form relationships with other adult females, usually called "aunts." Thus, when a bonnet infant loses its mother there is someone to turn to as long as its aunts remain; but when a pigtail infant loses its mother, it has no one to act as a substitute. When a bonnet infant is separated into a single cage, reports Kaufman (1973), it still does not become depressed, presumably because it has developed "sufficient security and coping ability to remain engaged with the environment, animate or inanimate, even in the absence of a mother or a substitute" (p. 47).

Separation environment. As might well be expected, the nature of the infant's environment during separation is also an important determinant of its response to separation. For example, the completeness of the separation clearly influences the response. Suomi, Collins, Harlow, and Ruppenthal (1976) found that

when mothers and infants were separated by wide-gauge mesh partitions, which allowed some limited physical contact between mother and infant, the reactions to separation were minimal. In a subsequent phase of the same study, the same mothers and infants were separated by a Plexiglas barrier, thus preventing any physical contact, and the usual protest–despair reaction ensued. Seay and Harlow (1965) found that infants separated by Plexiglas were more severely affected by separation than were infants whose mothers were housed in a different room, out of sight and hearing. Kaufman (1973, 1977) also found that pigtail macaque infants that had essentially recovered from the loss of the mother relapsed into despair when the mother was returned but made inaccessible for physical contact. So if physical contact is permitted, reaction to separation appears to be minimal; but if physical contact is not permitted, the "out of sight, out of mind" aphorism appears to hold true.

A second important environmental variable appears to be whether the separated infant also has to adjust to new surroundings. Hinde and Davies (1972) compared the effect of removing the mother, as opposed to removing the infant, from the home environment. They reported that rhesus infants were more severely affected during the separation period when they were removed than when their mothers were removed, in spite of the fact that the mother-removed rhesus infants were not adopted by the remaining members of the social group. Specifically, Hinde and Davies (1972) found that the pattern of the protest–despair response itself was quite different in the two conditions. Infant-removed monkeys tended to show a very prolonged protest phase characterized by high levels of vocalization and hyperactivity, and hence the despair phase was delayed in these monkeys. By contrast, the mother-removed infants showed only a very brief protest phase and quite rapidly fell into a state of despair. More recently, Hinde and McGinnis (1977) examined two more groups of rhesus infants reared in the same pens as those in the Hinde and Davies study. In one group the mother and infant were removed from the pen and housed together; in the other group the mother and infant were both separated from the group and housed individually. Hinde and McGinnis found that the infants in the former group showed only minimal disturbance, whereas those from the latter group were comparable to the infant-removed monkeys of the Hinde and Davies study (i.e., they exhibited a prolonged and intense protest phase before lapsing into despair). Thus, it seems that a prolonged protest phase is most likely to ensue when the separation occurs in a strange environment with no other available attachment objects.

This last point brings up a third important variable: the availability of substitute care during separation. As discussed earlier, bonnet infants that are readily adopted by "aunts" after separation do not exhibit depressive reactions, whereas pigtail infants that are not adopted by "aunts" become depressed (Kaufman & Rosenblum, 1969). Both rhesus and Java monkey infants react like pigtails when separated into an environment with an unfamiliar mother substitute—they are not adopted and they often become depressed (Schlottmann & Seay, 1972; Suomi & Harlow, 1977). In addition, Kaufman and Stynes (1978) recently found that a separated bonnet infant left in a group of pigtails became depressed in spite of Kaufman's (1973) earlier report that bonnets separated into single cages did not become depressed. In the more recent case, the depression may be attributable to being left in the familiar home environment with only unresponsive pigtails as alternative attachment objects. These findings resemble numerous reports in the human literature. Robertson and Robertson (1971) and Bowlby (1973) both contend that the provision of a mother substitute is one of the best ways to minimize the disruptive consequences of separation in human children.

As Bowlby (1973) also reported for human children, if substitute care in the form of a mother is not available, then substitute objects of attachment in the form of familiar peers can attenuate the response to separation. For example, Suomi et al. (1973) found that infants separated from their mothers and housed with a familiar peer showed less intense depressive symptoms than infants housed alone. These differences were exaggerated several months later. However, if the peers

are not familiar, the response may be the opposite. Lewis (1978) reared rhesus infants alone with their mothers and then separated them either into single cages or into peer groups. He found qualitatively more severe depression and withdrawal in the infants separated into a peer group than in those housed singly. Presumably because these infants had no social skills for interacting with peers before separation, they were less able to cope with a complex new social situation after separation than were infants living alone. It is as yet unclear how monkeys reared in a large group before separation (and hence more socially competent than Lewis's animals) would react to being placed into a group of unfamiliar peers. Furthermore, how this factor would interact with whether the separation environment were familiar or strange (cf. Hinde, 1977; Hinde & McGinnis, 1977) is also unclear.

In sum, one can see that separation environment variables can interact in a complex way with preseparation history to help determine how a given subject will respond to separation. It should be emphasized, however, that this interaction is neither simple nor as yet completely understood. It is clear that considerable work remains to be done in this area.

Peer Separation Studies

Basic findings. In recent years a number of investigations have been devoted to the study of social separation from nonmaternal objects of attachment. Infant monkeys are quite capable of forming strong attachment relationships with objects other than their mothers, for example, surrogates (H. F. Harow, 1958) and peers (H. F. Harlow, 1969), and it is now clear that separation from such attachment objects can produce full-fledged protest–despair reactions. Suomi, Harlow, and Domek (1970) first demonstrated that infants raised together since birth without their mothers exhibit the classic signs of protest and despair when separated from one another, starting at 90 days of age. Novak (1974) reported protest and despair following surrogate separation in monkeys reared from birth on the surrogates. Suomi et al. (1976) reported that qualitatively similar protest and despair reactions to maternal and peer separations can be elicited from the same subjects over a span of a few weeks. In their study monkeys were first separated from their mothers but allowed unlimited access to their peers; 2 weeks later they were also separated from their peers. Clear and qualitatively similar patterns of protest and despair were elicited by both the maternal and the peer separations. Most recently, Singh (1977) reported protest and despair in five rhesus infants raised in a feral environment when each was removed from the rest of the group for a 4-day period. They did not exhibit the dramatic depressive behaviors often seen in the laboratory, such as self-clasping and huddling, but definitely showed decreases in range of locomotion, appetite, levels of vocalization, and environmental exploration.

Factors Influencing Response to Peer Separation

Species, age, and sex. To date, peer separation studies have only been conducted systematically with rhesus monkeys, so it is impossible to assess whether there are any species differences. Sex differences appear to be nonsignificant in all published studies, and age does not appear to be a critical factor, at least within the first year of life (see below for a discussion of separation in older monkeys). However, the relative lack of effects disclosed may in part result from the paucity of studies concerned with these factors.

Duration and number of separations. No single study has systematically varied the duration of peer separations within or between subjects in order to assess the influence of this variable. In the published studies on separation, however, the duration has varied from 2 days (Erwin, Mobaldi, & Mitchell, 1971) to 4 weeks (Suomi & Harlow, 1975). Since the environments into which the infants are usually separated are generally single cages in peer separation studies, it is not surprising that little evidence of behavioral recovery is seen even over several weeks (e.g., Suomi, Seaman, Lewis, DeLizio, & McKinney, 1978).

Several studies have examined the effect of multiple peer separations. In the first such study, Suomi et al. (1970) separated monkeys that had been reared together in groups of four since birth. Between 3 and 9 months of age they were subjected to a total of 20

separations, each of 4 days' duration, with intervening 3-day reunion periods. During separation the monkeys were housed individually but could see and hear each other. The response to each separation was characterized by initial agitation (protest) followed by despairlike behaviors. The subjects showed no sign of adaptation to the separations, exhibiting as severe reactions to the 20th separation as to the 1st, except that protest was reduced at the end. Seaman (1975) subjected monkeys reared together since birth to three 19-day separations between the 4th and 10th month of life. These subjects also responded to each separation with protest and despair, although by the 3rd separation protest had diminished considerably.

Preseparation variables. Little attention has been devoted to the importance of preseparation variables for response to peer separation. One possible exception is Suomi (1976), who reported that the behavioral repertoires of subjects prior to separation were related to the severity of the response to separation. Specifically, those infants that had been raised with their mothers for the first 3 months of life and then put into peer groups showed less severe signs of despair than peer-only monkeys that were separated at 5–7 months of age. Similarly, Kraemer (1978) found peer separation responses to be far more pronounced in peer-reared subjects than in mother–peer-reared subjects. The subjects raised with their mothers did not have behaviors such as self-clasping and self-mouthing in their repertoires prior to separation, and they did not exhibit these behaviors after separation, as did the monkeys reared in peer groups since birth. The mother-reared infants did, however, show other less dramatic signs of despair—decreased play and activity levels as well as decreased appetite. Alternatively, the mother-reared monkeys may have shown less severe signs of despair because they were not as attached to their peers as were the peer-only monkeys rather than because of different behavioral repertoires per se. More recently, Singh (1977), too, has drawn attention to the importance of preseparation behaviors in determining the nature of response to separation. His feral monkeys had been raised in an environment in which they had to be constantly vigilant for predators. When separated from their peers they became depressed, yet they remained alert and vigilant and did not show the extreme depressive behaviors seen in the laboratory—self-clasping and huddling. These behaviors were less likely to appear not only because of the need to remain vigilant but also because they were not part of these monkeys' preseparation repertoires.

Separation environment. Two studies have directly examined the effects of the separation environment on response to separation. Suomi (1973) separated groups that had been raised together since birth either into single cages or into vertical chambers that restricted activity in addition to providing complete isolation from other monkeys. Suomi found that the monkeys that spent their separation period in the vertical chambers were more severely affected, both in the short term and in the long term, than monkeys that were separated into single cages. Suomi and Harlow (1975) reported essentially the same finding in another study in which only one monkey at a time was separated from the group, either into a single cage or into a vertical chamber. Those monkeys separated into single cages became less depressed, and their peer groups as a whole less disrupted, than the monkeys separated into the vertical chambers.

Singh (1977) included some incidental observations on the effects of the separation environment in his study. He noted that because each separated monkey was still living in its home environment while the rest of the group was removed, the depressive response was not as severe, presumably because of the need to remain vigilant for the danger of predators. He further noted that this situation contrasts sharply with the situation in which monkeys are separated into vertical chambers or single cages and have no need to remain vigilant.

As yet, little more is known about the effect of separation environment on response to peer separation. No study thus far has examined the effect of separating monkeys from one long-standing peer group into a new peer group—either one that is already established or one that is newly formed. Further work in this area should be of interest, given the evidence discussed earlier of the importance

of such variables in mother–infant separation studies.

Separation in Older Monkeys

Thus far we have only discussed separation studies performed on monkeys under 1 year of age. Older members of a species can also experience social separation, and recently a good deal of attention has been directed to what the nature of the response to separation may be in older monkeys. The majority of these studies have involved peer separation rather than maternal separation, for two reasons. First, under normal conditions (i.e., feral environments), monkey infants gradually separate themselves from their mothers (and vice versa); so by the time the monkey has reached adolescence, the strength of its attachment to the mother has weakened considerably (H. F. Harlow & M. K. Harlow, 1965; Kaufman & Rosenblum, 1967). Second, for practical reasons, most laboratories routinely separate offspring from their mothers long before the offspring have reached adolescence. Consequently, the majority of separation studies in adolescent and adult monkeys have involved peer separations.

Short-term peer separation in older monkeys. In general, studies involving short-term separation in adolescents have reported less severe reactions than for infants in comparable paradigms. For example, Bowden and McKinney (1972) separated recently formed pairs of peers, ranging in age from 2.5 to 4 years, for a 2-week period. They found signs of mild protest but no despair. Similarly, McKinney, Suomi, and Harlow (1972) repeatedly separated 3-year-old peers for four 2-week periods and found very mild reactions. These monkeys had been reared together since infancy but had also lived with their mothers for the first 8 months. In addition, Erwin, Brandt, and Mitchell (1973) separated 3-year-old heterosexual pairs of monkeys that had lived together for 1, 2, or 3 weeks and found some evidence of disturbance upon separation, but hardly a dramatic protest–despair response. Interestingly, the pairs that had been together the longest (3 weeks) were, if anything, *less* disturbed by the separation than were the other pairs. Maple, Risse, and Mitchell (1973) also reported little disturbance when recently formed (2 days) adult heterosexual pairs were separated.

Together, these four studies might be interpreted as suggesting that adolescents and adults are relatively immune to the devastating consequences of separation that younger subjects often exhibit. An equally plausible interpretation, however, may be that the monkeys in these studies did not exhibit severe reactions because they were not strongly attached to one another prior to separation. This interpretation receives some support from evidence, discussed below, which indicates that if adolescents are strongly attached to one another (e.g., when they have lived together since infancy without their mothers), then under certain conditions they do indeed exhibit dramatic protest–despair responses to peer separation.

Separations among nuclear-family-reared monkeys. The effects of separating 5-year-old monkeys that have been reared in complex social groups called nuclear families (M. K. Harlow, 1971) have recently been examined by removing them from their family environments (Suomi, Eisele, Grady, & Harlow, 1975). Each nuclear family group contained four mother–father pairs and their offspring, and so the offspring had access to other adults and to peers before separation. Subjects were removed from their nuclear families and subsequently housed with a group of familiar peers, with a group of strangers, or alone. The group-housed monkeys displayed only mild reactions to separation from their families, but the individually housed subjects all exhibited protest, followed by a deep depression. Therefore, under appropriate circumstances, older monkeys (a 5-year-old monkey is nearly a full-fledged adult) can exhibit severe responses to social separation.

Long-term repeated peer separations in adolescents. Very recently, two additional studies have been completed which indicate that peer separations can also have severe consequences for adolescents reared together in groups of four without mothers since infancy. Suomi, Mineka, and DeLizio (Note 1), in two independent experiments, subjected peer-reared rhesus monkeys to repeated peer separations (a total of 16 4-day separations in one study

and 24 4-day separations in the other) in the second or fourth year of life. During some of the separations, all of the subjects displayed the classic protest–despair reactions described earlier for younger monkeys. Most of the older monkeys had also been subjected to separations in the first year of life; these monkeys showed protest and despair on the first adolescent separation. One group of 4 subjects, however, had never been separated prior to adolescence, and they did not exhibit clear-cut biphasic reactions to the first few separations. However, from the third separation on, they exhibited as severe depressive reactions upon separation as did members of the other groups. In both studies, there was also a tendency for the monkeys' reactions to repeated separation to become more severe over time. The protest phase virtually disappeared after a few separations, and the monkeys began to drop immediately into despair. (See below for a more complete discussion of these results.)

Response to Reunion

Nature of the Response to Reunion

Bowlby (1960, 1973) observed that following a prolonged period of separation, reunion between the mother and the child was often marked by detachment, in which the young child expressed indifference and/or hostility toward his or her returning mother. There was a tendency for detachment to be more pronounced the longer the duration of the separation. Although clear-cut similarities between human children and monkey infants in terms of the protest–despair response to separation have been repeatedly demonstrated, no published monkey study has yet reported convincing evidence that detachment follows despair behaviors; indeed, a majority of studies have in fact found that attachment behaviors between the separated partners tend to increase over preseparation levels immediately upon reunion. For example, Seay et al. (1962) and Seay and Harlow (1965) reported that mother–infant pairs showed enhanced attraction to each other upon reunion, but after 3 weeks these mother–infant interactions had returned to baseline. Similarly, Hinde et al. (1966) and Kaufman and Rosenblum (1967) reported that reunion typically involved immediate increases in mother-directed contact-seeking and contact-maintaining behaviors and that after several weeks of reunion these behaviors largely returned to preseparation levels. In a more intensive and detailed analysis of reunion behaviors, Spencer–Booth and Hinde (1971) reported that during reunion infants not only spent more time on their mother and less time at a distance from her but also played a greater role in maintaining proximity to her than they did before separation. These temporary increases in attachment behavior during reunion have also been observed in feral mother–infant pairs (Singh, 1975) and in peer groups (e.g., Suomi et al., 1970). Both McKinney et al. (1972) and Bowden and McKinney (1972) witnessed extensive reciprocal mounting, mock aggression, and vigorous grooming following reunion of their juvenile peer-separated subjects.

There are several studies in the literature that have reported detachmentlike phenomena, but none of them seem to be describing quite the same thing that Bowlby reported in humans. For example, Erwin et al. (1971) reported that their juvenile pairs tended to ignore or avoid one another for the first few minutes of reunion. They interpreted this behavior as some evidence of detachment, but since it lasted for only a few minutes, it can hardly be said to be evidence of true detachment. In addition, Rosenblum and Kaufman (1968) reported that bonnet infants often spent the majority of their time with their adopted mothers following reunion with their biological mothers. One bonnet infant took a week to re-establish ventral contact with its biological mother, and one never did. However, it is not clear that this behavior should be defined as detachment since none of these bonnet infants ever showed despair during separation. Finally, Rosenblum (1971) and Kaufman (1973) reported that pigtail infants that had largely recovered from the severe phase of despair often relapsed into that state when their mothers were introduced into their cages in a restraining open-wire cage. This apparent despair upon seeing the mother could be considered analogous to detachment only if the same thing happened when the mother

was made accessible. As Kaufman recently found, however, this does not happen: Infants immediately respond when given the opportunity to press a bar to regain access to their previously caged mothers (Kaufman, Note 2).

Factors Influencing Response to Reunion

Species, age, and sex. Overall, monkeys' responses to reunion seem to be more consistent across a wide range of situations than are responses to separation. There are no reports of substantial species variation in the response to reunion that occurs in depressed monkeys, and there appear to be no obvious age differences in reunion, at least in the first year of life. Hinde and Spencer-Booth (1971) are the only investigators to have noted any sex differences. They found a tendency for males to spend less time off the mother than females and for males to initiate a smaller proportion of leavings. However, as was the case for responses to separation, these sex differences in reunion responses are in the same direction as those observed in nonseparated mother–infant pairs of the same age (White & Hinde, 1975); so whether they are truly sex differences in response to reunion per se is unclear.

Duration and number of separations. Hinde and Spencer-Booth (1971) compared the effects of a single 6-day separation, two 6-day separations, and one 13-day separation. They found no qualitative differences in behavior among these three groups during reunion. However, the infants that had undergone only a single 6-day separation appeared to return to preseparation activity levels faster than did the other two groups, thus suggesting that longer and more mother–infant separations may somewhat slow the process of recovery. However, no other studies have directly compared the effects of these variables on mother–infant reunion following separation.

Among the numerous studies performed on peer separations, not one has examined directly the effect of duration on response to reunion. Suomi et al. (1970) and McKinney et al. (1972) reported that the response to reunion did not change noticeably with a small number (up to 12) of repeated separation–reunion cycles. However, more recently, Suomi et al. (Note 1) found dramatic changes in reunion behavior with repeated separations in adolescent monkeys. In all the subjects examined, although despair reactions became more severe with repeated separations, the response to reunion dampened. In fact, distinctions between separation and reunion behavior patterns became increasingly blurred, with reunion behaviors clearly changing more in the direction of despair behaviors rather than vice versa. During later reunion periods these juveniles simply sat next to each other passively, often clutching their own bodies. The reciprocal mounting and mock aggression activities often characteristic of reunion in animals of this age disappeared across the repeated separations and reunions. (See below for a further discussion of these results.)

Preseparation history. A number of aspects of preseparation history have been shown to influence the subsequent response to reunion. For example, the duration of the attachment relationship interrupted by the separation may be important. Erwin et al. (1973) reported that the longer a heterosexual pair had been together before separation (1, 2, or 3 weeks), the less disturbed they were upon reunion and the quicker they returned to preseparation behaviors. Whether the length of the attachment over longer time spans also has an effect, and if it is in the same direction, is not known at this point.

Behavioral repertoires before separation are also clearly important. Erwin et al. (1971) noted that peer groups reacted differently upon reunion than did mother–infant groups. They observed a brief detachmentlike phase in the juvenile pairs that has not been observed in mother–infant studies. These differences may largely be due to the differences in preseparation behavioral repertoires of monkeys raised with mothers as opposed to with peers. Suomi (1976) also noted that monkeys that are less sophisticated socially before separation as a function of previous social history generally have a harder time readjusting during reunion. Moreover, Hinde and Spencer-Booth (1971) observed that the more tension in the mother–infant relationship before separation, the more likely the infant was to show severe distress upon reunion. Tension was indexed by the relative frequency of times the infant was rejected by the mother and by the

proportion of time the infant spent in trying to maintain proximity to the mother. High distress during reunion was indexed by less movement, more sitting, and more whoo-calling.

Separation environment. The characteristics of the separation environment may also have an influence on the response to reunion. Suomi (1973) found that peers separated into vertical chambers exhibited qualitatively different behaviors upon reunion than peers separated into single cages. The chambered monkeys tended to self-clasp more during reunion than the caged monkeys, which spent significantly more time clinging to one another. These results suggest that the more restrictive the separation environment, the longer it may take to reestablish an attachment relationship with a partner.

In a somewhat different vein, Hinde and Davies (1972) and Hinde and McGinnis (1977) reported that infants that had been removed from their social group during separation recovered faster during reunion than did infants whose mothers had been removed from the social group during separation, leaving the infants in the group cage. At first glance, this may seem surprising in light of the conclusion reached from Suomi's (1973) experiment, which suggested that the infants that had been removed would be more upset because their separation environment had been more restrictive. However, there is a straightforward resolution of this contradiction, which stems from the functional difference in reunion environments alluded to in the Hinde and Davies, and Hinde and McGinnis studies.

Reunion environment. These investigators (Hinde and Davies, 1972; Hinde and McGinnis, 1977) noted that when a mother was returned to her social group for reunion with her infant, she had some difficulty reintegrating herself into the group and thus had less time to devote to her infant. They argued that this probably accounted for their finding that the group of mother-removed infants as well as the group of mother- and infant-removed infants had a harder time recovering to preseparation levels of behavior than did the group of infant-removed infants, as indicated by the greater role that the former two groups of infants had to play in maintaining contact with and proximity to their mothers. According to these investigators, when the infant has been removed from the group instead of the mother, the mother has no readjustment problems herself upon reunion, and the relationship normalizes itself much faster. Their additional observation that mother-removed infants that had been depressed in the home pen continued to remain depressed for a considerable time after reunion seemed to be a result of their staying in the same environment; this depressive affect may have been conditioned to the environment (Gewirtz, 1961).

Further support for the viewpoint that differences in the behavior of the mothers influence recovery is supplied by Kaufman (1973), who compared the reactions of bonnet and pigtail mothers and infants during reunion. He reported that pigtail mothers were more protective and restrictive and that they discouraged their infants from straying far from the group, whereas the bonnet mothers were much less restrictive and let their infants play where they pleased. The consequence was that pigtail infants spent a greater amount of time in proximity to their mothers than before separation, and this increase was maintained for several months. Bonnet infants, by contrast, returned to preseparation behaviors immediately and continued on their normal course of social development.

Suomi and Harlow (1975) also showed that the characteristics of the peer group can influence response to reunion following peer separation. In their sequential separation study, one infant at a time was removed from the group and then returned. Each time a new infant came back depressed, the group behavior became less "normal", and it became harder for each subsequent infant to return to preseparation patterns of social interaction. A similar point has been made by Chappell and Meier (1975), who noted that the maturational arrest observed by Suomi et al. (1970) in peer groups that had undergone multiple separations may have been a function of the poor caretaking provided to each infant during reunion—caretaking was provided only by other infants that were equally stressed by separation. They suggested that infants multi-

ply separated from their mothers may not show such deleterious effects because their mothers could provide better caretaking during reunion periods. In Chappell and Meier's study, reunion behavior did tend to normalize faster after the fourth mother–infant separation than after the first, providing at least some support for this hypothesis regarding the results of the Suomi et al. (1970) study.

Long-Term Effects of Separation

General "personality" change. Although the majority of studies on mother–infant separation have reported that the enhanced attraction between mother and infant during reunion is only a temporary phenomenon, a number of investigators have found some long-term effects of early separation experience. Hinde and Spencer–Booth (1971) reported that the tendency of separated infants to try to maintain more contact with their mothers and to leave them less to interact with other conspecifics was still present 6 months following reunion. In some infants this tendency was detectable as long as 2 years following reunion. These findings are consistent with Bowlby's (1973) report of anxious attachment in human children who have undergone separation.

Mitchell, Harlow, Griffin and Møller (1967) reported that very short separations (2 hours) during the first year of life still had pronounced effects when the monkeys were 2 years old. These subjects were much more withdrawn when exposed to novel social stimuli such as unfamiliar adult monkeys than were same-aged controls that had not experienced the short separations during their first year. Thus, even separations that appear to be minor at the time may have long-term consequences. It should be noted, however, that these effects may not be specific to separation. Novak (1974) demonstrated that other nonseparation manipulations that elicit strong fear reactions in the first year of life produce long-term consequences similar to those reported by Mitchell et al. (1967). Accordingly, caution should be exercised in attributing long-term consequences to maternal separation per se as opposed to any one of a number of traumatic events.

Suomi et al. (1973) compared the long-term effects of separation into two different environments. In their study infants were permanently separated from their mothers at 2, 3, or 4 months of age and either housed with peers or in single cages. Over the next few months the peer-housed infants slowly began to recover, and they grew up to be fairly normal. The single-caged infants, by contrast, did not recover and still showed high levels of disturbance behaviors at 6 months of age.

In addition, Suomi et al. (1970) demonstrated that a series of 12 separation–reunion cycles in peer-raised monkeys had long-term consequences of maturational arrest. Behavior patterns such as ventral clinging and self-mouthing, which are generally absent in 1-year-old monkeys, were still at very high levels, and more advanced patterns of behavior such as environmental exploration and play were virtually absent in these monkeys.

Predisposition to Severe Separation Reactions Later in Life

In the human literature it has often been reported that individuals who have experienced traumatic social loss as children may be predisposed to respond to social loss as adults with a strong depressive reaction (Bowlby, 1973; Brown, Harris, & Copeland, 1977; Heinicke & Westheimer, 1966). For example, Brown et al. (1977) found that loss of a mother before age 11 is associated with greater risk of depression for adult women. They describe such early loss as creating a vulnerability to the effects of other types of loss or stress in adulthood. Actually, the evidence supporting this position is not unequivocal (e.g., Akiskal & McKinney, 1975). Brown and his colleagues do, however, seem to have made substantial progress in recent years in delineating the critical variables that determine whether early loss will create a predisposition to severe reactions later in life (e.g., Brown, Bhrolcháin, & Harris, 1975; Brown et al., 1977). The same hypothesis has also been investigated in primate studies, and again the evidence has been mixed, although some recent work of Suomi et al. (Note 1) has begun to delineate which factors are critical in creating this predisposition to severe reaction later in life.

Young, Suomi, Harlow, and McKinney (1973) subjected infant monkeys to a single 30-day peer separation and then housed them

in social units until they were 3 years old, at which time they were subjected to repeated daily separations (23 hours/day). The monkeys that had experienced separation early in life manifested many more depressive symptoms in reaction to the adolescent separations than did a control group that had experienced no previous separations. McKinney, Kliese, Suomi, and Moran (1973), by contrast, failed to find any reliable detrimental effects of early isolation on later separations at 3 years of age. There are two reasons, however, why the McKinney et al. (1973) study may not directly contradict the conclusions reached by Young et al. (1973). First, McKinney et al. housed their subjects in vertical chambers during separation, a manipulation known to have more adverse effects than the single cages that Young et al. used; thus, McKinney et al. may well have observed a ceiling effect that did not allow for any differences between previously isolated and nonisolated controls to emerge. Second, McKinney et al.'s subjects were quite different in their sociobehavioral repertoires at the time of the adolescent manipulations, making direct comparisons between the two groups' reactions to separation difficult.

Another study failing to find any reliable effect of early separation on sensitizing a monkey to react adversely to later separations is that of Suomi et al. (1976). In this study monkeys that had been subjected to maternal and peer separation(s) at 6 months were also subjected to permanent separation from their mothers at 1 year. Suomi et al. found no more adverse reactions in this group than in a control group that had undergone no previous separations. Again, however, there may have been a ceiling effect, because the previously nonseparated 1-year-old monkeys were quite adversely affected by this manipulation.

Most recently, Suomi et al. (Note 1) have found evidence that early experience with separation may predispose monkeys to react more severely than nonseparated monkeys only to the first in a series of adolescent separations. By the second or third separation, the previously nonseparated monkeys in their study were as depressed as the ones with a prior history of separations. These results seem to suggest that early separations may only lower the threshold for adverse reactions to the first of a series of later separations rather than create a general predisposition for especially adverse reactions to all such separations. Suomi et al. (Note 1) also found that an important factor in determining what effect early separation has on response to later separation is what the nature of the response to the early separation was. In their second experiment, monkeys that had responded in the first year of life with a severe despair response also showed a severe response when separated in their second year. By contrast, monkeys that had shown only a mild response to the early separations responded with only minimal despair to the later separations.

Theoretical Interpretations of Separation Phenomena

We have seen that a multitude of variables interact in a complex way to determine the form of an individual monkey's response to separation from its attachment figure(s). The response can vary from virtually no reaction to severe life-threatening depression. In recent years four prominent theories have attempted to explain the nature of the response to separation and to make some sense out of why particular variables influence that response in the way they do. In the following section we examine these four theories with regard to their ability to explain existing data, occasionally commenting on their usefulness in providing a predictive framework within which to generate future research hypotheses (see also Suomi, Mineka, & Harlow, in press). It is our contention that data collection in future primate separation studies could be usefully guided by one or more of these theories if they succeed in integrating in a reasonable fashion the existing data. We wish to emphasize at the outset that these theories are not mutually exclusive and often provide complementary perspectives on different aspects of the response to separation, namely, its evolutionary–adaptive function, its associated cognitive components, or its temporal dynamics. Hence no attempt will be made to support one theory at the expense of another; instead, each will be evaluated with respect to its ability to provide an explanatory and predictive framework for

that aspect of the separation reaction with which it is concerned.

The first two theories are concerned primarily with the adaptive nature of the response to separation. Both stem from a combination of the ethological and psychoanalytic traditions—Bowlby's (1969, 1973) attachment-object-loss theory and Kaufman's (1973, Kaufman and Rosenblum, 1967) conservation–withdrawal theory. The third theory —Seligman's (1974, 1975) learned helplessness approach to separation—deals primarily with the cognitive–motivational nature of the separation response. Finally, Solomon and Corbit's (1973, 1974) opponent-process approach addresses the changes in the dynamics of attachment and separation reactions that may occur over time.

Bowlby's Attachment-Object-Loss Approach to Separation Phenomena

Attachment, fear, and separation. Bowlby's (1973) treatment of separation phenomena derives from his earlier theoretical assumptions about the functions of attachment behavior. He has argued convincingly that one of the chief functions served by an attachment object, particularly a mother, is that of providing a base of security (Bowlby, 1969). From this security base the infant can explore the environment, and furthermore he or she can return to the attachment object when confronted with any one of a whole array of frightening stimuli. According to Bowlby, a child who is confronted with a frightening stimulus not only withdraws from that stimulus but also retreats toward an attachment object.

Given these assumptions about the function of attachment behavior, it follows that when the attachment object is gone, so also is the security base. When there is no longer refuge from frightening stimuli, the situation becomes much more frightening and sometimes terrifying. Naturally under such conditions, being separated from one's attachment object can be very unpleasant. Following this line of reasoning (and there is considerable evidence to support it in both the human and nonhuman primate literature), Bowlby has argued that there is a universal response to separation from an attachment object. This response is one of distress, or protest, and when the separation becomes extended, despair often ensues.

Separation anxiety and anxious attachment. An individual who has experienced the unpleasant experience of separation from an attachment object is not likely to forget it and may well develop a fear that at some time in the future separation will again occur. Once an individual develops a fear of separation, he or she is fairly likely to exhibit anxious attachment. An anxiously attached child is likely to spend excessive amounts of time and energy maintaining proximity to his or her attachment object and to be reluctant to leave its sight for even short periods of time. In addition, anxiously attached children appear to be more likely to display a severe reaction to subsequent separations than are securely attached individuals. Moreover, it should be noted that factors other than fear of separation, for example, constitutional factors or caretaking characteristics of the mother, may also contribute to anxious attachment.

Applications of Bowlby's approach to primate separation data. Generally speaking, Bowlby's (1969, 1973) theoretical views seem quite consistent with a large part of the primate separation data discussed earlier. First, the biphasic protest–despair response to separation that Robertson and Bowlby (1952) proposed long ago has been observed in many of the primate species studied. The detachment component of the separation reaction, however, does not appear to occur regularly, if at all, in nonhuman primates. Perhaps detachment requires much greater symbolic and cognitive representational abilities than exist in nonhuman primates.

A second contention of Bowlby's theory that is also consistent with numerous findings in the primate literature is that one of the main functions an attachment figure serves is to provide a security base. For example, H. F. Harlow and Zimmerman (1959) noted long ago that infant monkeys consistently retreat to a cloth surrogate mother when faced with novel or threatening stimuli, generally showing considerable relief and relaxation as they cling to their surrogate. By contrast, infant monkeys reared without a surrogate or with only a wire surrogate remain terrified in the presence of the same novel or threatening objects.

Third, Bowlby (1969, 1973) has predicted that separated individuals will react less adversely if they have access to substitute attachment figures during separation. Evidence supporting this contention comes from studies showing that monkeys living with familiar peers (e.g., Suomi et al., 1976) are less adversely affected by loss of their mothers than are monkeys living alone or with unfamiliar peers (Lewis, 1978). In addition, the pigtail–bonnet species differences in response to separation seem to be at least partially the result of a functional difference in the availability of "aunts" who will adopt the separated infants.

Fourth, Bowlby's prediction that anxiously attached infants will be the most severely affected by separation is consistent with the findings of Spencer–Booth and Hinde (1971). They showed that rhesus infants that spent the most time trying to maintain contact with the mother prior to separation also showed the most tension in the relationship upon reunion. Kaufman and Rosenblum (1967) have reported essentially the same finding for pigtail monkeys.

Finally, Bowlby's theory provides a viable explanation for why even brief separations such as those used by Mitchell et al. (1967) can produce such long-term consequences. Those short (less than 2-hour) separations were apparently sufficient to establish a fear of separation in the young rhesus subjects. Overall, then, Bowlby's approach provides a good theoretical framework within which one can integrate a great deal of existing data in the primate separation literature. It also seems to provide a useful framework for generating research ideas concerning factors that may attenuate or exaggerate the response to social separation as well as the effect that separation may have on the future dynamics of the attachment relationship.

Kaufman's Conservation–Withdrawal Theory

The conservation–withdrawal approach to separation of Kaufman (1973; Kaufman & Rosenblum, 1967) also stems from an evolutionary–functional tradition. Kaufman has suggested that separation is merely one of many different kinds of stressful situations that may elicit biphasic responses, such as protest and despair, in any given organism. For an organism faced with a stressful situation, there is first a fight–flight reaction, which may later be followed by a phase of conservation–withdrawal if the point is reached where the fight–flight response threatens the organism with exhaustion. The protest (agitation) that occurs upon separation is seen as an example of the fight–flight response and despair (depression) as an example of conservation–withdrawal. In addition to his emphasis on the neurobiologically adaptive nature of the separation response, Kaufman (1973, 1974) also stresses the cognitive–subjective aspects of the protest–despair response. In particular, he considers the protest phase to be best characterized as a state of anxiety, indexed by the high degree of agitation, vocalization and anguished facial expressions. The despair phase, indexed by withdrawal, inactivity, collapsed posture, and grieflike facial expressions, is thought by Kaufman to be characterized by feelings of helplessness and hence depression.

Kaufman (1973, Kaufman & Rosenblum, 1967) has argued that this particular biphasic response is very adaptive for a young infant that has lost its mother. First, the general agitation characteristics of protest, for example, excessive locomotion and vocalizations, may serve to draw attention to the separated infant and to increase the likelihood of its mother finding it or vice versa. However, this phase cannot last indefinitely without both exhausting the infant and drawing the attention of potential predators. Therefore, the phase of despair, characterized by inactivity and withdrawal, can be considered adaptive in that it serves to conserve resources and decrease the probability of detection by predators at a time when the probability of reunion is quite low. Kaufman emphasizes that despair is not caused by fatigue but occurs instead in order to prevent fatigue from setting in. This latter point is supported by the observation that the infant can easily mobilize its resources if a danger situation arises (e.g. Singh, 1977) and can readjust to normal activity levels immediately after reunion with its mother.

Kaufman's theory is also consistent with a number of neurobiological approaches that

view the nervous system as organized to mediate two opposing responses to stressful situations (e.g., Engel, 1962; Hess, 1954; Riss & Scalia, 1967). Furthermore, the physiological correlates of protest and despair discussed earlier appear to be relatively consistent with the biphasic response theory of the neurological approaches. Thus, from an evolutionary and physiological standpoint, Kaufman's theory provides a logical analysis of the protest–despair response to separation, with empirical support for a number of its contentions. The theory seems to be most useful in its predictions regarding the physiological correlates of the protest–despair response, especially how they resemble other biphasic responses to different kinds of stressors. The cognitive–subjective correlates of the reaction to social separation have not yet been adequately operationalized by Kaufman to be of great predictive utility (but see below for how Seligman has attempted to operationalize them).

Learned Helplessness Approach to Separation

Helplessness, depression, and separation-induced despair. Recently, Seligman and his colleagues (Seligman, 1974, 1975; Seligman, Klein, & Miller, 1976) have proposed that it is the perception of helplessness that underlies the despair–depression response to separation. This idea is part of Seligman's learned helplessness theory of depression, which states that despair or depression occurs when an organism perceives reinforcement and relief to occur independently of its responding, namely, when it perceives that it is helpless to control the environment. Seligman (1975) has suggested that separation from an attachment object is one very effective way of inducing such a sense of helplessness, because the attachment object is the major figure through which an organism gains a sense of control over its environment.

The helplessness theory of depression and separation-induced despair is primarily a cognitive theory. When an organism perceives that it has no control over important life events, cognitive, motivational, and affective deficits are presumed to ensue: The organism will be retarded at learning in future situations in which it lacks control, its motivation to respond will be reduced because of a belief that its responses will be ineffective, and the affect of the individual will be characterized by sadness, passivity, and depression. Thus, this theory predicts that cognitive, motivational, and affective deficits should all occur in separation-induced depression. The human literature clearly documents that cognitive and motivational deficits do occur in children who have undergone social separation (e.g., Bowlby, 1973; Spitz, 1946), although recent evidence seems to suggest that the cognitive deficits are more a result of the deprived environment that children are often separated into than a result of the separation from the attachment figure per se (see Rutter, 1972, for extensive reviews of this topic). So far, no direct tests of this hypothesis have been made in studies of separation in nonhuman primate subjects. Certainly the symptoms of despair, such as low activity, loss of appetite, and greatly reduced play and aggression behaviors, suggest that a variety of motivational deficits do exist in separated primate subjects. However, there is as yet no evidence bearing on the question of whether a separated organism is retarded at learning that its responses can bring gratification or relief and that it has control over its attachment object. In addition, no direct studies have addressed whether general cognitive deficits ensue following separation, and if they do, to what extent they are the result of environmental factors as opposed to separation per se.

The helplessness hypothesis also makes interesting predictions regarding variables that should attenuate the response to separation. When an individual loses a salient portion of its environment because of separation and no functional substitutes are available, it will have lost a good deal of control over its social options and opportunities. Moreover, if the separation environment is different from that of preseparation, the individual's behavioral repertoire may not be appropriate, and the perception of loss of control will be magnified. By contrast, if substitute attachment objects are available or if other means of exerting control still exist in the environment, the response to separation should be greatly attenuated. As documented in the review above, these situations seem to be ones in which separation is least likely to elicit a strong

despair response. For example, monkeys with the most sophisticated behavioral repertoires prior to separation (e.g., bonnet macaques) or with functional, substitute attachment objects during separation are not likely to exhibit strong depressive symptoms. Yet if put into an environment in which their behavioral repertoire is inadequate to exert control (e.g., with a group of pigtails), bonnet infants may become depressed (Kaufman & Stynes, 1978). The helplessness hypothesis also predicts that individuals having control over reunion or individuals that are separated into an environment that still contains multiple means of exerting control should also show an attenuated despair reaction. There are as yet no data bearing on these hypotheses.

The helplessness approach to separation-induced despair also identifies a number of factors that may be involved in predisposing an organism to a particularly severe, or a particularly mild, response to separation. Specifically, this hypothesis states that previous experience with uncontrollable traumatic events such as separations should predispose an individual to react adversely to subsequent separations. Conversely, a previous history of mastery or control over important life events should "immunize" the organism from the debilitating effects of separation. The evidence reviewed earlier that bears at least indirectly on these hypotheses is at best mixed. Monkeys with an early history of separation may or may not be especially predisposed to an adverse reaction to separation as an adolescent or adult (McKinney et al., 1973; Suomi et al., 1976; Suomi et al., Note 1; Young et al., 1973). The Suomi et al. (Note 1) results discussed earlier are consistent with helplessness theory predictions insofar as the monkeys that weathered early separations well did not show a severe despair response to the later separations, whereas the monkeys that displayed dramatic despair responses to early separations—perhaps indicative of a sense of helplessness—also responded badly to the later separations. Unfortunately at this point the factors mediating the differential response to early separations remain unidentified. No study has yet directly manipulated the controllability of an organism's environment to determine its effect on the organism's reaction to subsequent separation. It is true that more socially sophisticated monkeys are somewhat less susceptible to separation-induced despair than are monkeys raised in more restricted social environments; this may be because they have had more experience in controlling their environment. However, nuclear-family-reared monkeys have also had a very rich environment in which to gain a sense of control, yet they are susceptible to separation-induced despair. Thus, the evidence supporting these hypotheses regarding predisposition is clearly mixed and must await the results of studies designed to test them directly.

Loss of control and separation-induced protest. The helplessness hypothesis discussed thus far has been concerned only with why an individual might exhibit a despair–depression response to separation. However, as discussed earlier, the initial reaction to separation from an attachment object is almost invariably protest, not despair. A recent integration of reactance theory with helplessness theory suggests one ready explanation of the biphasic nature of protest–despair responses. Wortman and Brehm (1975) have argued that when an organism loses control and senses that its freedom is being threatened, its initial reaction is to become aroused and attempt to restore its freedom (reactance). The degree of reactance is a function of both the importance of the freedom being threatened and the organism's expectation that it possesses freedom to being with. When an organism is separated from its attachment object, it clearly loses the freedom to interact with that attachment figure. One can see that the agitated behavior that occurs during protest following separation is consistent with Wortman and Brehm's description of reactance.

Wortman and Brehm (1975) have also proposed that the reactance phase does not persist indefinitely. With prolonged exposure to uncontrollable outcomes, the organism will finally give up trying to restore its freedom, namely, it will become helpless in Seligman's (1975) terms. Roth and Kubal (1975) have reported support for this integration of reactance theory and helplessness theory for humans in an experiment in which they found that brief treatment with uncontrollable outcomes produced reactance, whereas pro-

longed treatment resulted in helplessness. If separation from an attachment figure is seen as an aversive uncontrollable event, then the initial response should be reactance (protest), which should ultimately turn into helplessness (despair) as attempts to restore the attachment figure fail. With repeated separations the reactance phase should be bypassed, and the organism should drop immediately into a phase of helplessness or despair. This is exactly the pattern of results seen in the Suomi et al. (Note 1) studies.

In sum, we can see that the helplessness approach to separation generates many interesting hypotheses to guide future research on social separation in primates. The theory is particularly useful in that it provides an operational framework by which to study the nature of the cognitive and motivational deficits associated with the response to separation. In addition, the theory makes interesting and useful predictions regarding what factors may attenuate or exacerbate the response to separation and what factors may predispose an organism to react with a mild as opposed to a severe depressive response to separations later in life. The results of studies designed to test such hypotheses will be of interest because they may provide useful information concerning the role of cognitive–perceptual variables in mediating the response to separation.

Solomon and Corbit's Opponent-Process Approach to Separation

A fourth approach to the understanding of social separation stems from Solomon and Corbit's (1973, 1974; Solomon, 1977) opponent-process theory of acquired motivational states. This theory uses the dynamics of drug addiction as a model for the acquisition of a whole range of complex motivations, including the attraction of dangerous sports, imprinting in precocial birds, and, for the purposes of this review, love–attachment relationships in humans and primates. According to the theory, the grief and despair exhibited by an individual following separation from an attachment figure are simply a natural reaction characteristic of a normally functioning, healthy nervous system. From a functional standpoint, the observed despair is analogous to the withdrawal symptoms shown by a heroin addict when he or she is separated from the drug. This viewpoint differs fundamentally from the medical model of both depression and drug addition, which assumes underlying pathology. As can be seen below, the opponent-process approach to separation is of particular interest and importance because, being a theory of the temporal dynamics of affect, it makes interesting predictions regarding changes in attachment and separation behaviors that may occur over time.

Opponent-process theory. There are three main propositions of the theory that are important for the present discussion. First, when a nervous system is stimulated by any intense, affect-arousing stimulus (such as exposure to an attachment object or an addicting drug), there occurs a primary affective reaction (*a* process) that has behavioral, subjective, and physiological components. Once the *a* process has been activated, a second process is automatically set into play by central feedback mechanisms; its exclusive function is to bring the organism back to hedonic neutrality. This second process (*b* process) is conceived as being opposite in hedonic quality from the *a* process, and it acts as an opponent process, diminishing the intensity of the *a* process. As the stimulus eliciting the *a* process disappears or dies out, the *b* process manifests itself in its full intensity for the first time and takes the organism on a hedonic excursion of its own. The *b* process dies out gradually. The result of the additive relationship between the *a* and *b* processes is as follows: When the stimulus first comes on there is a peak of affective arousal, which is followed by a phase of adaptation as the *b* process automatically comes into play and subtracts from the intensity of the *a* process. State A is a combination of *a* and *b* processes and is less intense than the initial peak of A. State B only manifests itself when the *a* stimulus is gone and the *b* process is no longer opposed by the *a* process.

The second major proposition of Solomon and Corbit's (1973, 1974) theory is that the process is strengthened with use. Its latency to recruit becomes shorter, and it becomes more intense and long lasting. Like any good unconditioned response, the *a* process, by contrast, does not change with repeated

stimulation. Therefore, there are dramatic changes in the dynamics of State A (manifest combination of the *a* and *b* processes) after prolonged use: The peak of A is no longer as great because the *b* process begins to subtract from A very quickly, the steady level of State A is much less intense than with early stimulations, and the intensity of State B when the *a* stimulus is gone is much more intense and long lasting. A third basic tenet of opponent-process theory is that States A and B, like any powerful hedonic states, are presumed to be conditionable to neutral cues (conditioned stimuli—CS_As and CS_Bs).

When applied to love–attachment relationships, this theory argues that prolonged or repeated exposure to an attachment object results in the strengthening of a *b* process whose function is to bring the organism back to hedonic neutrality. Dynamically speaking, this means that after prolonged exposure to the attachment figure, the attachment appears less intense due to a dampening of State A by the growing *b* process. However, at the same time, the potential for distress and despair upon separation from the attachment figure (State B) becomes much greater, also because of the growing *b* process. Environmental stimuli (CSs) that have been consistently paired with either State A or State B have the capacity to modulate whatever state the organism is in.

Repeated separation in monkeys from an opponent-process perspective. Two recent studies outlined earlier have been interpreted in an opponent-process framework (Suomi et al., Note 1). In these studies a total of four groups of monkeys (three or four per group) were subjected to two or three series of eight separation–reunion cycles (4 days of separation plus 3 days of reunion) in their second or fourth year of life. Three of the groups had previous experience with separation in their first year; the fourth group did not. Except for during previous separations, all members of a group had lived together since approximately 15 days of age.

Four major patterns of results emerged from these studies, each of which fits with predictions of opponent-process theory, assuming that State A exists when the animals are together and State B when they are apart. First, the theory predicts that State B (the state of despair) upon separation should get stronger with repeated separations because of a growing *b* process. Consistent with this prediction, Suomi et al. (Note 1) found that across each series of separations the monkeys spent more and more time engaging in self-directed behaviors indicative of severe despair. Concurrently, they spent less and less time engaged in environmentally directed behaviors across each series of separations. These trends are particularly striking since developmental changes occurring during the same age period are in the opposite direction: Normal monkeys exhibit less passive and self-directed behavior and more locomotion and exploration at these ages (H. F. Harlow, 1969; H. F. Harlow & M. K. Harlow, 1965). As these despair behaviors became more dramatic across repeated separations, the protest behaviors that normally occur immediately following separation dropped out. If the protest response is seen as a by-product of any major frustration or disruption in the environment, then one would expect it to habituate or to be bypassed for a faster drop into the state of despair, State B (e.g., Amsel, 1967; Wortman & Brehm, 1975). In other words, protest may not be an intrinsic part of the *b* process but rather may just be a manifestation of the disruption that occurs upon separation. (See Hinde and McGinnis, 1977, for results at least partially consistent with this interpretation.)

Second, the changes in behavior over the course of each reunion were also consistent with predictions of opponent-process theory. Specifically, the theory predicts that State A will peak right after reunion followed by a decline in the intensity of A with time as the *b* process has a chance to recruit. The results of the Suomi et al. (Note 1) studies indicated that levels of social activity indicative of attachment were much higher on the first than on the third day of reunion. Concurrently, nonsocial activities showed an increase from Day 1 to Day 3.

Third, the changes in the absolute levels of social behavior during reunion across repeated separation–reunion cycles were also consistent with predictions of the theory. In particular, the theory predicts that the absolute intensity of State A should decline with

repeated separation–reunion cycles because of the growth of the *b* process. The results of Suomi et al. (Note 1) indicated dramatic declines in social activities over repeated separation–reunion cycles at least on the third day of reunion and for some subjects on the first day. At the same time, there were increases in nonsocial environmental activities. By the end of each separation series, on the last reunion day, the levels of social activities in animals of this age were extremely low. Furthermore, the social activity they did show was largely passive in nature, unlike the play, aggression, or sex behaviors characteristic of normal monkeys of this age. Yet despite this appearance of minimal attachment to one another after these repeated separations, this was precisely the time when the monkeys showed the most dramatic and immediate despair upon separation, as predicted by opponent-process theory.

Finally, age differences in the degree of despair behavior exhibited by the different groups were also consistent with opponent-process predictions. According to the theory, the monkeys that lived together longer (4 vs. 2 years) should show a more dramatic despair response upon separation because they should have a stronger *b* process. In keeping with this prediction, Suomi et al. (Note 1) found that the 4-year-old monkeys exhibited levels of despair or depression that were considerably more dramatic than those seen in the 2-year-old monkeys.

Overall, the opponent-process approach to attachment and separation in primates appears to provide a useful predictive framework within which to study dynamic changes in attachment and response to separation. Because it is a theory of the temporal dynamics of affect, it makes specific testable predictions about changes that occur over time in response to separation and reunion. Furthermore, opponent-process theory provides an excellent framework within which to tease apart the relative contributions of associative and nonassociative factors to phenomena associated with attachment and separation. In this regard, it should be noted that what the relative contributions of associative and nonassociative processes to the observed changes in behavior discussed above were is not yet clear. For example, our earlier discussion implied that nonassociative variables such as the automatic strengthening of the *b* process through repeated separation were the major contributors to the observed changes. However, it is likely that associative variables also made important contributions to the observed changes. In particular, acquired anticipations or expectations—CS_As and CS_Bs in opponent-process terms—are likely to have played a role in modulating States A and B. For instance, trained observers present during separation may have become CS_Bs. As such, because they were also present during reunion, they may also have contributed to the dampening of State A, observed over repeated reunions and indexed by decreases in social activities. Because these studies were not originally designed with opponent-process theory in mind, these variables cannot as yet be teased apart. However, opponent-process theory appears to provide a framework within which such variables can be teased apart in future research.

Summary and Conclusions

In this article we have reviewed the present state of knowledge regarding social separation in nonhuman primates. As in human children and across a wide range of primate species and situations, there is frequently a biphasic protest–despair response to separation from an attachment figure. There is also, however, enormous individual variation in response to separation. As with human children, the most important determinants of this response appear to be preseparation history and the nature of the separation environment. Those monkeys that are anxiously attached prior to separation and the monkeys that are separated into an environment with no functional substitute attachment objects available are the individuals most likely to show a profound despair reaction. With respect to reunion with the attachment object following separation, the same set of variables seems to be important in determining how quickly behaviors will return to their preseparation levels. Generally during reunion there is an initial increase in attachment behaviors that gradually subsides back to preseparation levels. A true analog to the detachment observed in some human children has not been reported

in nonhuman primates. Long-term effects of separation (e.g., anxious attachment) have been observed in some monkeys several years later, but it is not clear how specific these consequences are to a history of separation, as opposed to any one of a number of traumatic events. Likewise, the data concerning whether early separations predispose an individual to respond adversely to later separations are mixed.

We have also reviewed briefly four current theoretical approaches to the understanding of separation. Both Bowlby (1973) and Kaufman and Rosenblum (1967) have proposed theories that stem from evolutionary–functional and psychodynamic traditions. Bowlby's attachment-object-loss approach stresses that there is a universal response of distress, and ultimately despair, upon separation from an attachment figure. This response will be attenuated if the individual is not anxiously attached or if there are functional attachment substitutes available. Kaufman proposes that the biphasic protest–despair response to separation is just one of many biphasic responses to stressful situations. Protest is an example of a fight–flight response and despair an example of conservation–withdrawal. Seligman (1974, 1975) has addressed the question of what the nature of the cognitions are in an individual that has been separated from its attachment figure. He suggests that it is a perception of helplessness that underlies and produces the cognitive, motivational, and affective deficits seen in separated, depressed individuals. Solomon and Corbit (1973, 1974) have proposed that their opponent-process model of the temporal dynamics of affect and acquired motivation provides a framework within which to study changes that occur over time in the dynamics of attachment and separation behaviors. Like Bowlby and Kaufman, they view despair reactions as natural responses that occur upon separation from an attachment figure. However, they also stress how these responses are likely to change over time.

Overall, it is apparent that future work on separation can usefully be studied from a number of different theoretical perspectives. In general, these different theories are not antithetical to one another but rather provide complementary information and ideas. It is likely that future research designed from all these approaches will be of value in furthering our understanding of the dynamics of both attachment and separation.

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